

USDA/EGRET model

30 September 2026

Overall question: How does temperature affect germination?

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- ▶ What are the average overall responses to chilling and forcing?
- ▶ And how does it vary by species? (And what is λ ?)
- ▶ And dormancy classes?

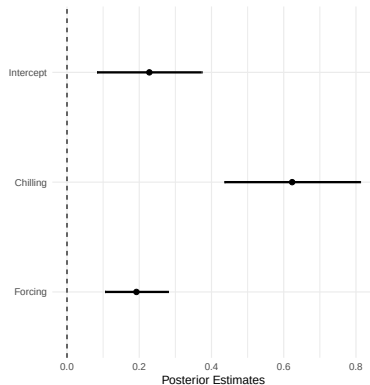
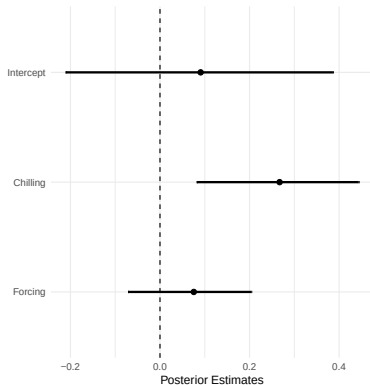
How the model works

$$p(\mu_i|\theta) = \begin{cases} 1 - \text{logit}^{-1}(\mu_i - k_1) & \text{if } \mu_i = 0 \\ [\text{logit}^{-1}(\mu_i - k_1) - \text{logit}^{-1}(\mu_i - k_2)] \\ \quad \times \text{Beta_proportion}(\text{logit}^{-1}(\mu_i), k_1) & \text{if } \mu_i \in (0, 1) \\ \text{logit}^{-1}(\mu_i - k_2) & \text{if } \mu_i = 1 \end{cases}$$

$$\mu_i = \alpha_{[sp]} + \beta_{f[sp]} \cdot \text{forcing} + \beta_{cs[sp]} \cdot \text{chilling}$$

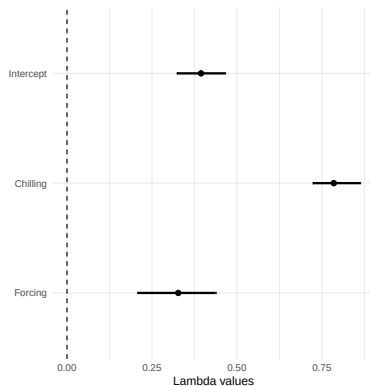
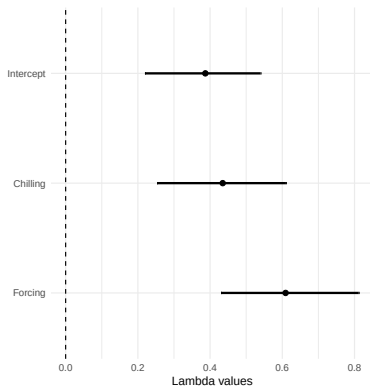
Results so far ... chilling/forcing

Overall: Gymnosperms vs. angiosperms



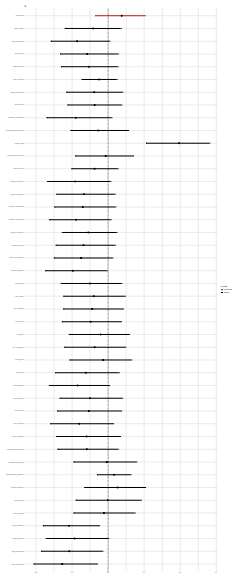
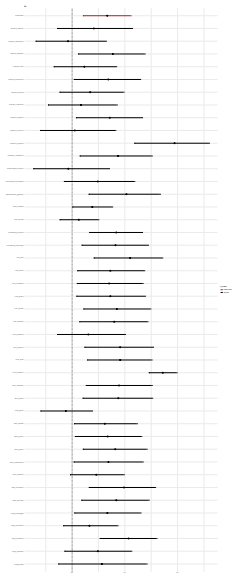
Results so far ... chilling/forcing

Phylogenetic effects: intercept, chilling, forcing (Gymnosperms vs. angiosperms)



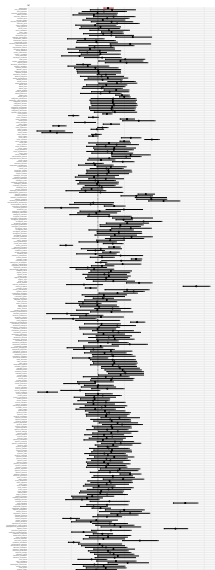
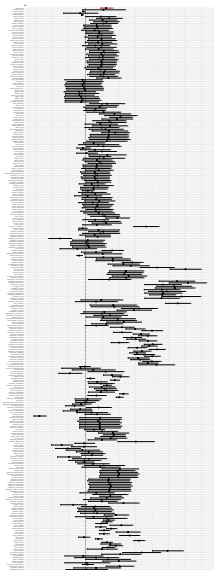
Results so far ... chilling/forcing

Individual species (Gymnosperms)



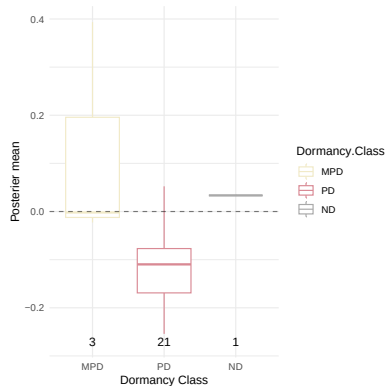
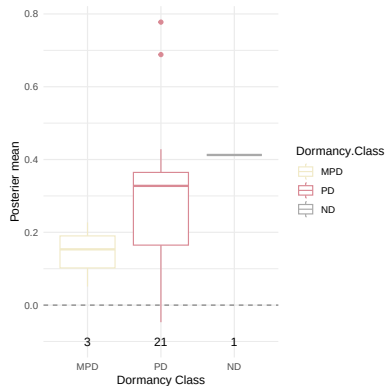
Results so far ... chilling/forcing

Individual species (Angiosperms)



Results so far ... chilling/forcing

Dormancy classes (Gymnosperms)



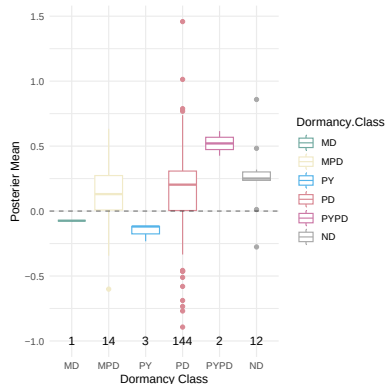
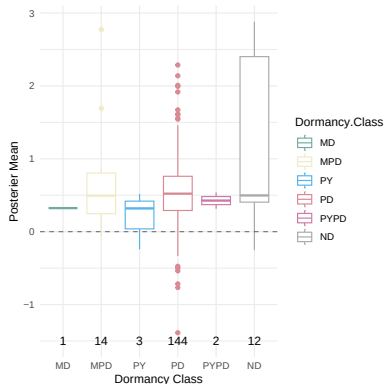
MPD: morphophysiological dormancy

PD: physiological dormancy

ND: No dormancy

Results so far ... chilling/forcing

Dormancy classes (Angiosperms)



MD: morphological dormancy

MPD: morphophysiological dormancy

PY: physical dormancy

PD: physiological dormancy

PYPD: combined dormancy ND: No dormancy

Other things we should look at?

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- ▶ Seed size?